

Sample Questions and Answers from Midterm Review

Q1: What is the loss function used in linear regression?

A1: The loss function used in linear regression is the Mean Squared Error (MSE), which is calculated as:

$$L(f) = (1/n) * \sum((y(i) - f(x(i)))^2)$$

Q2: What are the key components of a convolutional neural network (CNN)?

A2: The key components of a CNN are:

- Convolutional layers
- Pooling layers (e.g., max pooling)
- Fully connected layers
- Flattening operation

Q3: How does backpropagation work in neural networks?

A3: Backpropagation works by computing the gradient of the loss function with respect to each weight in the network using the chain rule, and updating weights to minimize the loss.

Q4: What is the function of the hidden state in RNNs?

A4: The hidden state in RNNs stores information from previous time steps in a sequence, allowing the network to maintain context over time and process sequential data.

Q5: What is the difference between a bi-directional RNN and a regular RNN?

A5: A regular RNN processes input sequences in one direction (typically forward). A bi-directional RNN processes the sequence in both forward and backward directions, which helps capture context from both past and future states.

Q6: When is a bi-directional RNN used?

A6: Bi-directional RNNs are used in tasks where context from both directions is important, such as in

speech recognition, machine translation, and named entity recognition.

Q7: What is overfitting and how can it be fixed?

A7: Overfitting occurs when a model learns the training data too well, including its noise, and performs poorly on new data. It can be fixed using techniques such as:

- Regularization (L1/L2)
- Dropout
- Early stopping
- Increasing training data
- Using simpler models